Software Test Plan (STP) - ATM System (Sample)

Project: Healthcare Appointment & Patient Record System

Version: 1.0

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Date: 11-10-2026

Status: Draft

1. Introduction

Purpose: This document defines the testing plan for the Healthcare Appointment & Patient Record System v1.0.

Scope: Testing covers login, patient management, doctor management, appointment booking, medical records, prescriptions, and security. Third-party services and infrastructure internals are excluded.

References: SRS v1.0, Software Architecture and Design Specification v1.0, RTM.

Definitions: STP – Software Test Plan; SRS – Software Requirements Specification; SAD – Software Architecture and Design; RTM – Requirements Traceability Matrix; RBAC – Role-Based Access Control.

2. Test Items

- Authentication module
- Patient management module
- Doctor management module
- Appointment management module
- Medical records module
- Prescription module
- Administration and access-control module

3. Features to be Tested

Requirement ID	Feature
HCS-F-001	User login
HCS-F-004	Patient registration
HCS-F-006	Doctor profile management
HCS-F-007	Doctor availability
HCS-F-008	Appointment booking
HCS-F-009	Appointment conflict prevention
HCS-F-010	Appointment rescheduling
HCS-F-011	Appointment cancellation
HCS-F-013	Medical history management
HCS-F-015	Prescription management
HCS-SR-002	Role-based access control
HCS-NF-001	Response time

4. Features Not to be Tested

- External service internals not developed by the project team.
- Third-party infrastructure and hosting internals.
- Features not included in the approved project scope.

5. Test Approach / Strategy

Levels: Unit testing, integration testing, system testing, and user acceptance testing.

Types: Functional testing, regression testing, performance testing, usability testing, and security testing.

Entry Criteria: Stable application build, test data available, and test environment ready.

Exit Criteria: All planned test cases executed, no critical defects open, and all acceptance criteria satisfied.

5.1 Security Validation

- Verify login and password handling.
- Check role-based permissions.
- Verify HTTPS/TLS configuration.
- Test unauthorized access to patient records.
- Check that sensitive information is not exposed in logs.

6. Test Environment

Hardware: Computer or laptop with network access.
Software: Next.js application, MongoDB database, and supported web browser.
Tools: Visual Studio Code, Postman, MongoDB Compass, browser developer tools, and Git.
Test Data: Sample patients, doctors, appointments, medical records, and prescriptions using dummy information.

7. Test Schedule

Milestone	Planned Date
Test case design	To be decided
Test environment setup	To be decided
Test execution start	To be decided
Test execution end	To be decided
User acceptance testing	To be decided

8. Test Deliverables

Software Test Plan

Test cases

Test scripts

Test data

Test execution results

Defect reports

Final test summary report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	**Sanketh Shivaji R**	Prepare plan, coordinate execution
Test Engineer	**Santhosh M**	Design & execute test cases, log defects
Developer	**Sairam Charan N**	Support defect fixes and triage
Product Owner	**Sumukh D Hegde**	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
Delayed application build	Begin test planning early
Database connection failure	Verify database setup before testing
Invalid test data	Prepare and validate sample data
Unauthorized data access	Test access controls for every role

11. Assumptions & Dependencies

- The application will be available for testing.

- MongoDB will be configured and accessible.
- Test accounts and dummy data will be available.
- Required features will be implemented before testing.

12. Suspension & Resumption Criteria

Suspend testing if:

- The application or database is unavailable.
- Critical defects block major test cases.
- The test environment becomes unstable.

Resume testing when:

- Blocking defects are fixed.
- The application and database are stable.
- Test data and environment are available.

13. Test Case Management & Traceability

The RTM maps requirements to test cases.

- HCS-F-001 (Login) → TC-AUTH-01, TC-AUTH-02
- HCS-F-008 (Book Appointment) → TC-APT-01, TC-APT-02
- HCS-F-009 (Prevent Conflicts) → TC-APT-03
- HCS-F-013 (Medical History) → TC-MR-01
- HCS-SR-002 (Role-Based Access) → TC-SEC-01

14. Test Metrics & Reporting

Metrics: Test cases executed, pass/fail percentage, defect count, defect age, and requirement coverage.

Reports: Test execution status and final test summary report.

15. Approvals

Role	Name	Signature / Date
QA Lead	**Sanketh Shivaji R**	11-10-2026
Dev Lead	**Sairam Charan N**	11-10-2026
Product Owner	**Sumukh D Hegde**	11-10-2026